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Research Article

Life Cycle Assessment for Spent Lithium-Ion Batteries' Recycling Process: Environmental Impact, Energy Consumption, and Sensitivity Analysis

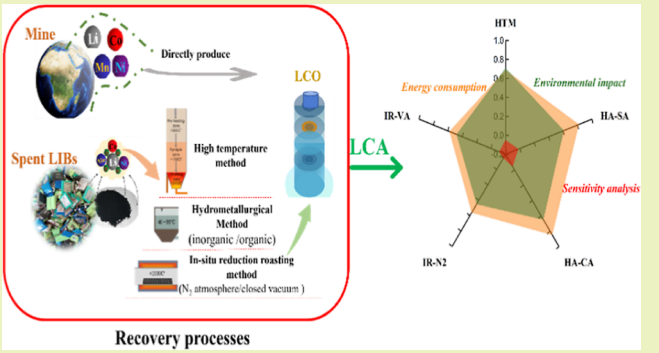
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Cite This: *ACS Sustainable Chem. Eng.* 2024, 12, 12966−12975

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ABSTRACT: The management of end-of-life lithium-ion batteries (LIBs) is a significant challenge for recyclers due to the increasing prevalence of electric vehicles. Considerable endeavors have been performed to advance the management of spent LIBs by means of the innovation and implementation of recycling techniques, including high-temperature and hydrometallurgical methods (sulfuric acid and citric acid), as well as the in situ reduction roasting method (vacuum and N_2). Nevertheless, the recycling process can potentially have adverse impacts on the environment and result in significant energy consumption, although the available data on this subject are currently limited. The present study utilized life cycle assessment (LCA) to comprehensively evaluate the environmental impact and energy consumption of these recycling processes. By conducting a comprehensive analysis of five distinct recycling systems, we have identified the crucial connections and primary elements that contribute to the mitigation of environmental consequences. While compared to the manufacture of new raw materials in industry, all recycling technologies have specific energy-saving and emission reduction impacts. The rates of consumption reduction are 68, 61, 38, 84, and 88%. This study provides a theoretical framework for firms and governments to establish more effective strategies for recycling of spent LIBs.



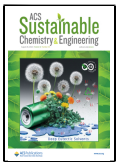
KEYWORDS: *life cycle assessment, spent lithium-ion batteries, recovery technologies, environmental impact, energy consumption, sensitivity analysis*

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INTRODUCTION

Lithium-ion batteries (LIBs) find extensive utilization in electronic devices and energy storage systems due to their exceptional energy density, lightweight composition, and portable nature. LIBs possess the advantage of being devoid of heavy metals such as mercury, cadmium, and lead, thereby minimizing pollution.[1] Additionally, LIBs exhibit commendable recycling capabilities, further enhancing their appeal in various applications. In the preceding three decades, there has been a notable upswing in the utilization of LIBs as an environmentally conscious energy solution for electric vehicles. Nations worldwide have taken proactive measures to champion the advancement of alternative energy automobiles and have put forth proposals for the substitution of conventional fuel-dependent vehicles. Prominent Chinese energy enterprises, such as BYD, have divulged a comprehensive schedule delineating their ambitious pursuit of complete electrification.[2] During the period spanning 2013 to 2020, China experienced a notable surge in both the production and the annual capacity of LIBs. In a notable development, the production of LIBs experienced a substantial surge, escalating from 4.8 billion units to an impressive 18.8 billion units.[3] Simultaneously, the worldwide demand for LIBs for electric vehicles was 120 GW h in 2019, and it is projected to increase to 680 GW h in 2025 and 1525 GW h in 2030.[4] Anticipated are noteworthy advancements in the development of novel energy vehicles accompanied by a corresponding surge in the requisition for LIBs as the primary means of propulsion. Nevertheless, it is imperative to guarantee the secure operation of electric vehicles that LIBs are substituted once their capacity diminishes below the threshold of 80%.[5] These batteries typically exhibit an average lifespan ranging from 3 to 5 years. It is anticipated that by the year 2030, the cumulative volume of spent LIBs on a global scale will ascend to an estimated magnitude of 11 million metric tons.[3] Therefore, the effective administration of spent LIBs is poised to assume an increasing significance in the imminent future.

Received: June 3, 2024
Revised: July 30, 2024
Accepted: July 31, 2024
Published: August 12, 2024



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https://doi.org/10.1021/acssuschemeng.4c04541
ACS Sustainable Chem. Eng. 2024, 12, 12966−12975

LIBs consist of four primary components: the cathode material, the anode material, the separator, and the electrolyte. Exporting cathode materials is expensive, with expected expenses for this component of LIBs reaching around 15 billion USD in 2022.[6] Scare metals like Co, Li, and Mn used in cathode materials are limited because of the industrial oligopoly. The leading two counties account for 88 wt % of Co and 79 wt % of Li production.[6] In 2019, almost 60% of the global Co supply originated from Congo, and the price of this metal experienced short-term fluctuations.[6] The projected demand for Co and Li in 2030 is six times and eight times more than that in 2020, with the price of USD 80,000 per ton and 16,500 per ton in 2018, respectively.[6] The limited production of rare metals and growing demand for them create a supply risk. If these valuable metals are recycled, they can transform waste into wealth and generate significant economic benefits.[7] Moreover, it is composed of elements possessing the potential to impose detrimental effects upon the surrounding ecosystem. The inadequate disposal of spent LIBs can lead to the emission of various substances, which possess the capacity to induce substantial environmental contamination.[8] The electrolyte $LiPF_6$, possesses the inherent capacity to engage in a chemical reaction with ambient water, leading to the liberation of HF gas.[8] It is worth noting that the presence of heavy metal ions within the cathode material has the potential to undergo dissociation in response to alterations in pH levels.[9] Consequently, this phenomenon carries substantial implications for the surrounding environment as it poses a considerable environmental hazard. The recycling of batteries and its consequential environmental ramifications have garnered significant attention and scrutiny since the 1990s.[10] In the study conducted by Halog et al.,[10] a comprehensive analysis was undertaken to compare the potential environmental ramifications associated with four distinct battery technologies, including lead−acid (Pb−A), nickel−cadmium (Ni−Cd), nickel metal hydride (NiMH), and sodium−sulfur (Na−S). Additionally, the researchers also conducted an assessment pertaining to recycling aspects of these battery technologies. It has been determined that Ni−Cd and Pb−A batteries are widely regarded as the most preferable options as they exhibit greater environmental compatibility. Nevertheless, it is crucial to prioritize the recycling of Ni−Cd and Pb−A batteries due to the inherent toxicity associated with the presence of metals, such as cadmium and lead.

The life cycle assessment (LCA) methodology is widely employed as a means of evaluating the ecological ramifications associated with a given product, process, or system over the entirety of its life cycle.[11] The utilization of this methodology enables the comprehension of the environmental implications of various procedures, determines the most favorable procedures, and identifies essential process components that may be amenable to potential enhancement. Currently, the application of LCA methodology has been employed in a systematic examination of various techniques employed for the recycling of batteries. Dunn et al.[12] undertook a comprehensive assessment utilizing life cycle and economic analysis to scrutinize five distinct variants of cathode material employed in LIBs. These materials encompassed $LiCoO_2$ (LCO), $LiMn_2O_4$ (LMO), $LiNi_{0.4}Co_{0.2}Mn_{0.4}O_2$ (NMC), $LiFePO_4$ (LFP), and $0.5Li_2MnO_3{\cdot}0.5LiNi_{0.44}Co_{0.25}Mn_{0.31}O_2$ (LMR-NMC). These researchers astutely observed that the cathode materials, contingent upon their specific composition, contribute to a substantial portion of energy consumption, ranging from 10 to 50%. In order to discern the pivotal phases that contribute to the overarching environmental load of LIBs, Tao et al.[3] employed LCA as a methodology to scrutinize the life cycle of LFP, encompassing their production, utilization, and recovery stages. The researchers further conducted a comparative analysis of various recovery technologies, ultimately concluding that the cascade utilization technique emerges as the most optimal choice. Quan et al.[13] undertook a comprehensive examination of diverse recovery methodologies pertaining to LFP batteries. Their discerning analysis led them to the resolute conclusion that hydrometallurgy, as opposed to pyrometallurgy and direct physical recycling, emerged as the preferred approach for the recycling of discarded LFP batteries. This preference was primarily predicated upon the superior environmental benefits afforded by hydrometallurgy. Jiang et al.[14] conducted an empirical investigation of NMC and LFP to ascertain the primary factors influencing the overall environmental performance. Their findings revealed that the material extraction and processing stages and the subsequent utilization stages emerged as the principal contributors to the aforementioned performance. It is worth noting that the environmental ramifications associated with the production and transportation phases of batteries are relatively inconsequential. However, it is crucial to acknowledge that the recycling of batteries holds immense potential in terms of augmenting environmental advantages.

In material recovery, the cathode material is the most concerning component due to its high contents of valuable metals. Various cathode materials have been produced, such as $LiCoO_2$ (LCO), $LiMn_2O_4$ (LMO), $LiNi_xCo_yMn_zO_2$ (NMC), LFP, etc.[6] The variations among various categories of spent LIBs are shown in Table S1. The major contents in different types of LIBs are shown in Table S2. The application of the recycling process for spent LIBs is determined by the physical and chemical characteristics of the valuable components within the battery. This process involves separating, extracting, and recovering. The conventional metallurgical processes, such as pyrometallurgy and hydrometallurgy, have been thoroughly investigated. At high temperatures, the metals suffer from reduction in the environment, resulting in the formation of an alloy. The alloy is then recovered. Pyrometallurgy is the method used for the extraction of metals. Hydrometallurgy involves the dissolution of metal components using acid or alkali solvents, followed by the separation of metals through extraction or chemical precipitation. Hydrometallurgy is a field of study that focuses on the process of separating metals. Consequently, the approach of recycling spent LIBs usually mixes the various processes of pyrometallurgy and hydrometallurgy.

In order to tackle the matter of recycling of depleted LIBs, numerous companies also have embraced either a pyrometallurgical or hydrometallurgical methodology. These approaches stem from the realms of Co or Li extraction metallurgy and have gained considerable traction. Notably, pyrometallurgical methodologies such as the Umicore, Inmetco, Accurec, and Glencore processes have successfully attained commercialization on an industrial magnitude.[15] However, it is imperative to acknowledge that the primary limitations of these technologies lie in their substantial energy demands and the consequential release of perilous gas emissions. Regarding hydrometallurgical techniques like GEM High-Tech, Brunp, Retriev, and Recupyl processes, a substantial quantity of leachates is necessitated to achieve optimal leaching efficiency.[15] This presents a significant hurdle in the realm of wastewater treatment and recovery of waste acid. In order to address these challenges, a number of innovative techniques have emerged, including the high-

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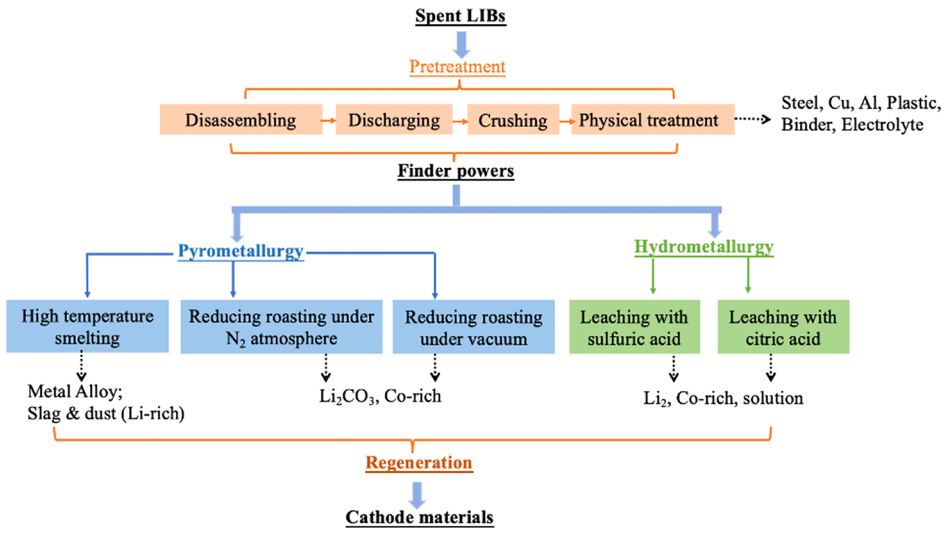


Figure 1. System boundary of spent LIB recycling.

temperature method, hydrometallurgical approaches employing organic or inorganic acid leaching, and the in situ reduction roasting method. Nevertheless, the majority of research endeavors are primarily focused on the advancement of processing techniques and the enhancement of operational parameters on a small-scale experimental level. Regrettably, practical implementation of these findings in large-scale industrial settings remains conspicuously lacking. While certain commendable evaluations have offered astute recommendations and directional growth prospects pertaining to the recycling of spent LIBs, there remains a notable dearth in the realm of comprehensive quantitative estimation of these nascent techniques in their industrial implementation. Consequently, a significant disparity persists between scholarly inquiry and practical realization of commercial viability. Therefore, conducting a comprehensive and systematic quantitative evaluation of these methodologies is crucial for advancing the progression and acceptance of recycling technology for spent LIBs.

In this study, an industrial recovery system containing a variety of new recovery methods was constructed by integrating existing studies of spent LIBs. The present study aimed to quantify the environmental impact and energy consumption of five different recovery methods employed in the whole life cycle of high-temperature metallurgy, the hydrometallurgical method of inorganics and organics, the inert atmosphere reduction roasting method, and the closed vacuum reduction roasting method. The quantification was carried out within the boundary of the LCA system. The study demonstrated the influence of modifying crucial key parameters in the system on its output. Furthermore, an analysis was performed to assess the variation among the various recovery methods, with the objective of identifying the most effective recovery strategy.

MATERIALS AND METHODS

Spent LIB Recycling Chain. The cathode material of spent LIBs in this study was identified as LCO. The process flow diagram for the spent LIBs' recycling chain assumed in this study, shown in Figure 1, is divided into the pretreatment phase, metal transformation, and regeneration stages. The pretreatment stage consists of disassembling, discharging, crushing, and physical treatment processes; herein, the plastic, electrolyte, and heavy metals (steel, Cu, Al, etc.) are liberated. Next, the finder powers (cathode material) that remain after separation are refined through a sequence of steps that comprise precious metal recovery, high-temperature smelting, reducing roasting, and targeted metal leaching. Li and Co are recovered first. Then, smelting is utilized to extract and purify metals of Li and Co for cathode material production. The liquid fraction from the leaching processes is assumed to be treated in industrial effluent plants. The residuals generated from the leaching process are assumed to be discarded in a landfill. The nonmetals isolated in the separation stage are not further processed or accounted for in this analysis, but they could be reused as raw materials or additives in new products such as wood−plastic composites and nonmetallic plates.

Goal and Scope. The goal of this study is to quantitatively assess environmental burdens and resource consumption associated with a formal recycling chain for spent LIBs, as implemented in China, including pilot pretreatment lines, metal transformation, and metal regeneration. The present investigation focuses on the utilization of five techniques for separating and recovering precious metals. These techniques consist of the high-temperature method (HTM), hydrometallurgical with sulfuric acid (HY-SA) method, hydrometallurgical with citric acid (HY-CA) method, in situ reduction roasting under a N_2 atmosphere (IR-N_2), and in situ reduction roasting under closed vacuum (IR-VA). Due to the significant amount of data involved in all stages of the life cycle analysis and experimental research of each recovery technique, it was not possible to obtain precise process data. Therefore, the current study depends on the following assumptions: (1) the composite potency of the electrode's active material, acquired post-treatment, is free of any contaminants; (2) the utilization of all materials is predicated on the stoichiometric coefficient, irrespective of any material wastage or reaction heat during the procedure.

The system boundary includes the steps from material liberation and separation through metal refining, as shown in Figure 1. Only the direct process steps themselves were considered. The functional unit was established as 1 t of spent LIBs. The composition of typical spent LIBs is specified in Table 1. The cathode material of spent LIBs was identified as LCO, and the content of the main metals in LCO is detailed in Table 2.

Table 1. Composition of Spent LIBs[3]

components	proportion (wt %)	components	proportion (wt %)
cathode material	29	electrolyte	10.8
carbon	18	plastic	4.2
copper foil	11	thermal insulation	0.37
aluminum foil	20	anode material	16.5
electronic part	1.4	aluminum shell	8.9

https://doi.org/10.1021/acssuschemeng.4c04541
ACS Sustainable Chem. Eng. 2024, 12, 12966−12975

Table 2. Composition of Main Metals in LCO[3,16]

components	proportion (wt %)	components	proportion (wt %)
Co	17.3	Al	5.2
Li	2.0	Fe	16.5
Cu	7.3	Ni	1.2
aluminum foil	20	anode material	16.5
electronic part	1.4	aluminum shell	8.9

Life Cycle Inventory Analysis. The description of the five scenarios is shown in S1. The input and output data for the processes in the recycling chain were determined by inventory analysis, as shown in Figure 2. The summary of conditions for recycling of Li and Co is shown in Tables S3 and S4.

In the pretreatment stage, the recovery efficiencies for Li and Co were 95% with a purity of 90%.[17−19] The production rate for the liberation and separation processes was 600 kg/h.[17] In the metal transformation stage, the leaching efficiencies for Li and Co were 97 and 98%, respectively; the recovery efficiencies for Li and Co were 95 and 97%, respectively.[20−23] The electrolytic reduction efficiencies were 100%. All metals recovered from the smelting recovery and leaching processes were pure metals.

Life Cycle Impact Assessment. The LCA step was carried out by using Simapro 9.4.0.1 software. The CML-IA baseline model was used to evaluate the environmental impact of the processes in the spent LIBs' recycling chain. In this work, the data on the energy flow and material flow for different recycling methods were mainly derived from our experimental results, patents, academic literature, and similar industrial processes.[17−23] The ecoinvent v3.5 database or theoretical calculations are due to the limited industrial data arising from the immaturity of the methods investigated in this study. Data sources of unit production energy consumption of various raw materials were obtained from Simapro 9.4.0.1, which is shown in Table S5.

The energy consumption of recycling technology is significantly influenced by the processing capacity, technological level, and energy structure. This study examines various scenarios of changes in the processing capacity, technological level, and energy structure from 2010 to 2040. The IR-N_2 process route is used as an example to compare and assess the changes in energy consumption of spent LIBs in recycling technology. The comprehensive analysis is shown in S2.

Evaluation Parameter. The life cycle effects of each recycling process were evaluated from environmental impact, energy consumption, and sensitivity analysis. A total of 10 indicators were selected to assess the environmental impact: abiotic depletion, global warming, ozone layer depletion, human toxicity, freshwater aquatic ecotoxicity, marine aquatic ecotoxicity, terrestrial ecotoxicity, photochemical oxidation, acidification, and eutrophication. The evaluation model of each indicator comes directly from Simapro version 9.4.0.1 software.

The reduction rate of energy consumption of various methods was defined based on eq 1.

$$\xi_E = \frac{E_v - E_r}{E_V} \quad (1)$$

where ξ_E is the consumption reduction rate, E_v is the energy consumption of new raw materials in industry, and E_r is the energy consumption of the recycling process.

RESULTS AND DISCUSSION

Environmental Impact Potential. Figure 3 illustrates the environmental implications throughout the life of the cycle of various recycling technologies. Among the several recycling methods available, the IR-VA and IR-N_2 recycling technologies have been identified as having the least environmental effects across their entire life cycles. The environmental repercussions of the HA-SA and HA-CA recycling technologies exhibit notable similarities, particularly with regard to their effects on eutrophication potential and acidification potential. The HTM recycling technology demonstrates a marginally superior environmental performance compared to the hydrometallurgical recycling method. However, it is important to note that the HTM recycling technology exhibits higher index values in

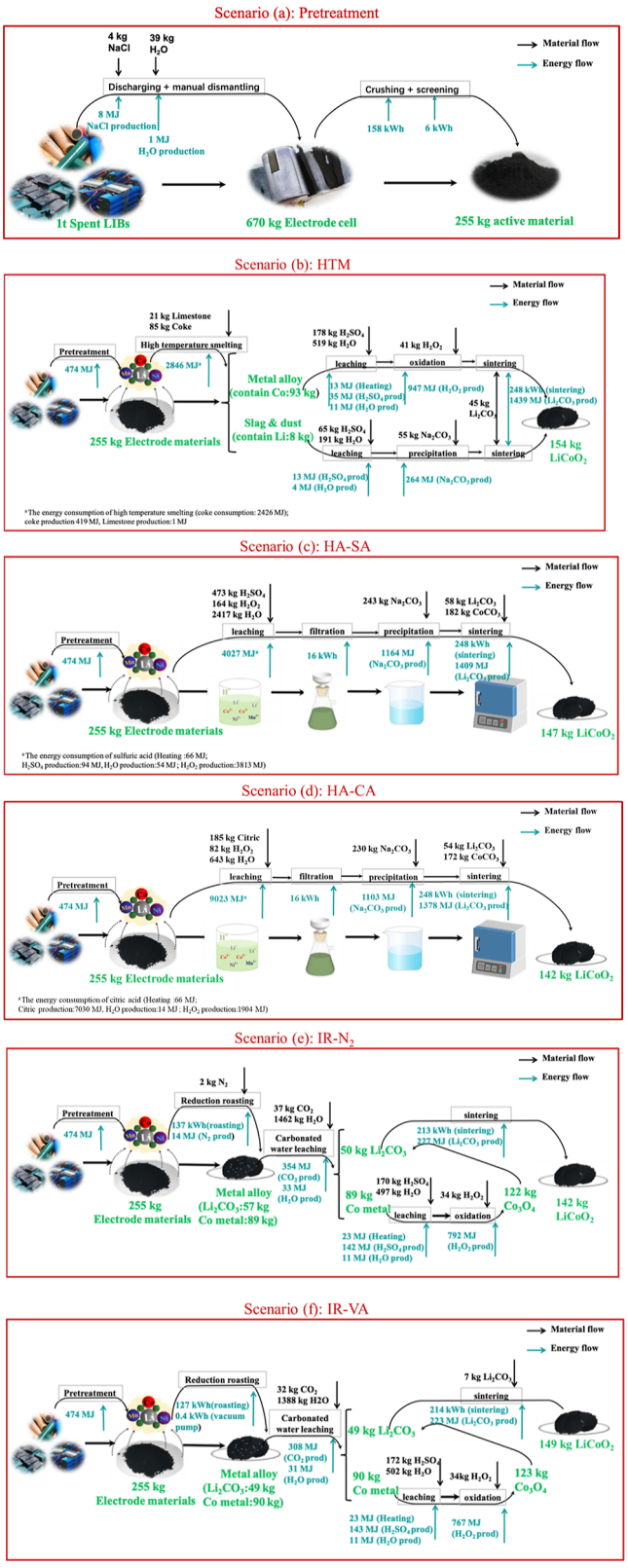


Figure 2. Input and output data for the recycling scenario: (a) pretreatment; (b) HTM; (c) HY-SA; (d) HA-CA; (e) IR-N_2; (f) IR-VA.

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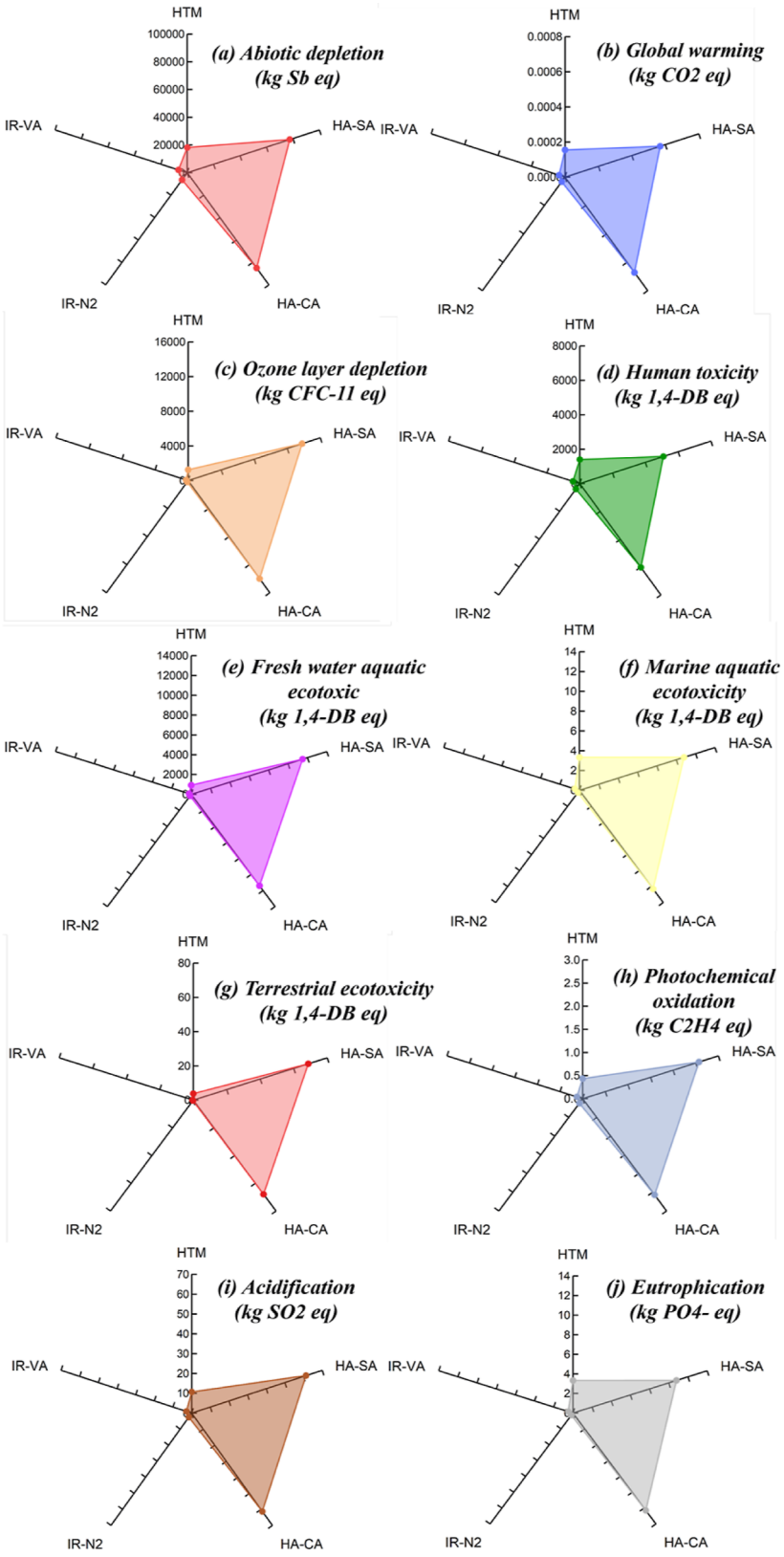


Figure 3. Environmental impact potentials in the life cycle of different recycling technologies.

relation to global warming, marine aquatic ecotoxicity, and eutrophication. This is due to the impact of various factors during the high-temperature smelting process, such as the slagging agent and direct emissions. Conversely, the hydrometallurgical recycling technology exhibits the most significant environmental impacts across its life cycle when compared to alternative technologies, particularly in terms of eutrophication potential and abiotic-resource depletion potential.

The environmental impact potential productions of different recycling technologies are shown in Figure 4. For the potential of

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Figure 4. Environmental impact potential productions of different recycling technologies.

abiotic depletion in the recycling process, HA-SA and HA-CA have the highest index values. The acid leaching process is primarily facilitated by sulfuric acid in HA-SA, while in HA-CA, it is facilitated by citric acid and sodium carbonate production. Conversely, IR-VA and IR-N_2 demonstrate a low index value. Electricity contributes the largest proportion to the potential in both IR-VA and IR-N_2. For the global warming potential in the recycling process, HA-SA and HA-CA exhibit the highest index values, effectively contributed by the sulfuric acid, cobalt carbonate, and citric acid production. IR-VA and IR-N_2 are found to have a low index value. Electricity contributes the largest proportion to the potential in IR-N_2. For the ozone layer depletion potential in the recycling process, the results were similar to those of global warming potential. It is worth noting that the production of sodium carbonate contributes the largest proportion to the potential of ozone layer depletion in HA-SA. For the human toxicity and the freshwater aquatic ecotoxic potential in the recycling process, HA-SA and HA-CA demonstrate the highest index values, effectively contributed by the sodium carbonate and cobalt carbonate production. For

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the marine aquatic ecotoxicity potential in the recycling process, HA-SA and HA-CA also exhibit the highest index values, while IR-VA and IR-N_2 demonstrate the lowest potential. The main contribution link of HA-CA is the citric acid production and heat consumption. For the terrestrial ecotoxicity and photochemical oxidation potential in the recycling process, HTM and HA-CA exhibit the highest index values while IR-N_2 demonstrates the lowest potential. The main contribution link of the HTM is diesel consumption. The production of citric acid contributes the largest proportion to the potentials in HA-CA. For the acidification potential in the recycling process, HTM and HA-CA exhibit the highest index values, while IR-VA demonstrates the lowest potential. This is due to in situ reduction roasting technology being a highly efficient method for breaking down organic materials at elevated temperatures in the absence of oxygen and without the need for added chemicals. For the eutrophication potential in the recycling process, HTM and HA-CA exhibit the highest index values while IR-VA and IR-N_2 demonstrate the lowest potential. The main contribution link of the HTM is diesel consumption. The production of citric acid and sodium carbonate contributes the largest proportion to the potentials in HA-CA.

Energy Consumption Analysis. The energy consumption for the production of 1 t of LCO amounts to 140 MJ.[24] The mass of LCO obtained from 1 t of spent LIBs ($LiCoO_3$) through five different recycling techniques is reported as 154, 147, 142, 149, and 142 kg, respectively, as shown in Figure 5.[24] The energy consumption values for the aforementioned methods are as follow: 21,560, 20,580, 19,880, 20,860, and 19,880 MJ/t, respectively. According to Figure 5, the energy consumption associated with the recycling of 1 t of spent LIBs ($LiCoO_3$) is significant compared to that involved in the industrial production of an equivalent quantity of LCO utilizing virgin materials. The HM-CA method demonstrates the highest total energy consumption, with a calculated value of 12,928.4 MJ. Additionally, its energy consumption per unit mass is determined to be 91.05 MJ/kg. IR-N_2 exhibited a minimum total energy consumption of 3209 MJ, with a corresponding energy consumption per unit mass of 21.54 MJ/kg. The energy consumption reduction rates of the HTM, HY-SA, HY-CA, IR-N_2, and IR-VA were 68, 61, 38, 88, and 84%, respectively. The results indicate that recycling spent LIBs has a higher potential to decrease consumption in comparison to the production of new raw materials in the industrial sector.

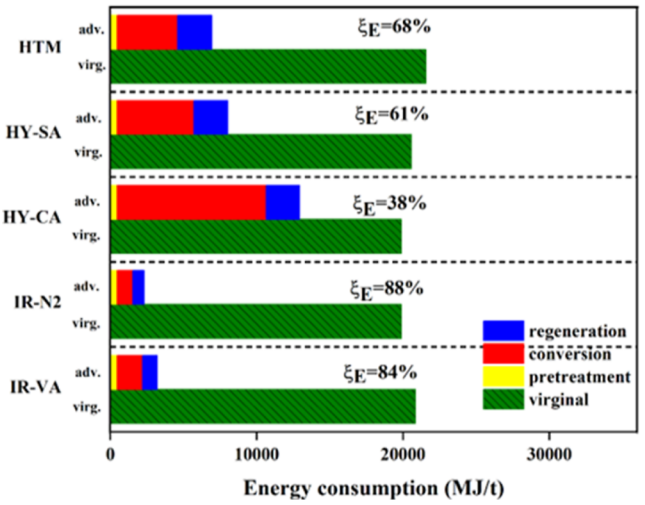


Figure 5. Energy consumption of the five LIB recycling processes.

The influence of different recycling techniques on the overall energy consumption can be observed in Figure 6. The energy consumption proportions of each process in HTM, IR-N_2, and IR-VA exhibit a highly significant amount of similarity. In the scenario of a HTM, the predominant determinant of energy consumption is the utilization of coke. Conversely, when considering the scenario of IR-N_2 and IR-VA, both roasting techniques demonstrate reduced operating temperatures. The sintering process is identified as the main factor contributing to energy usage, constituting 26.8 and 28.2% of the respective techniques. In the scenario of HY-SA and HY-CA, the principal determinant influencing the production procedure pertains to the production of H_2O_2 and citric acid, constituting 50.5 and 56.5% of the total, respectively.

The energy consumption of spent LIB recycling technology will vary based on treatment technology under varying processing capacity, technological level, and energy structure. As depicted in Figure 7, the technological integration exhibits the most significant reduction in consumption rate compared with all other scenarios, with a decrease of 38%. This is followed by the processing capacity, which experiences a reduction of 23%. Lastly, the energy structure shows a reduction of 12%. The energy consumption of the processing capacity continuously reduces over time. This is a result of the growing quantity of discarded batteries and rising energy consumption associated with their recycling process. The energy structure's rate of consumption decrease is the lowest, with minimal fluctuations over time, consistently remaining at approximately 12%. The energy consumption of technical integration gradually rises and eventually stabilizes. In 2030, the number of spent LIBs destroyed in China will have climbed dramatically. The progress in technology will facilitate a higher quantity of these spent batteries to be recycled over their lifespan.

Sensitivity Analysis. The results of a sensitivity analysis are presented in Figure 8, which was conducted to assess the impact of changes in particular variables on energy consumption and global warming potential. The baseline for this analysis is a 70% recovery rate.[24] The results illustrate that variations in the rate of recovery exert a substantial influence on the energy consumption and global warming potential associated with each method. In relation to energy consumption (Figure 8a), the HY-SA and HN-CA methods exhibited a considerable degree of variability, ranging from 44 to 45%. The second method utilized for reduction calcination involved the use of IR-N_2 and IR-VA, resulting in a variation range of 15−14%. The HTM demonstrated a notable level of stability in relation to variation in the recovery rate, with the range of sensitivity spanning from 6 to 7%. Figure 8b illustrates the influence of modifications in Li and Co recovery on the global warming potential associated with each corresponding method. The results reveal that the global warming potential of the HY-SA and HY-CA methods exhibits a strong sensitivity to changes in the recovery rate, potentially resulting in a fluctuation of up to 16 and 19%, respectively. The global warming potential of the alternative method did not exhibit sensitivity to changes in the recovery rate, showing a narrow range of variation of merely 5%.

DISCUSSION

Pretreatment Process. The energy consumption associated with the pretreatment processing of spent LIBs is relatively insignificant, ranging from 8.2% (HM-CA) to 13.9% (HTM) of

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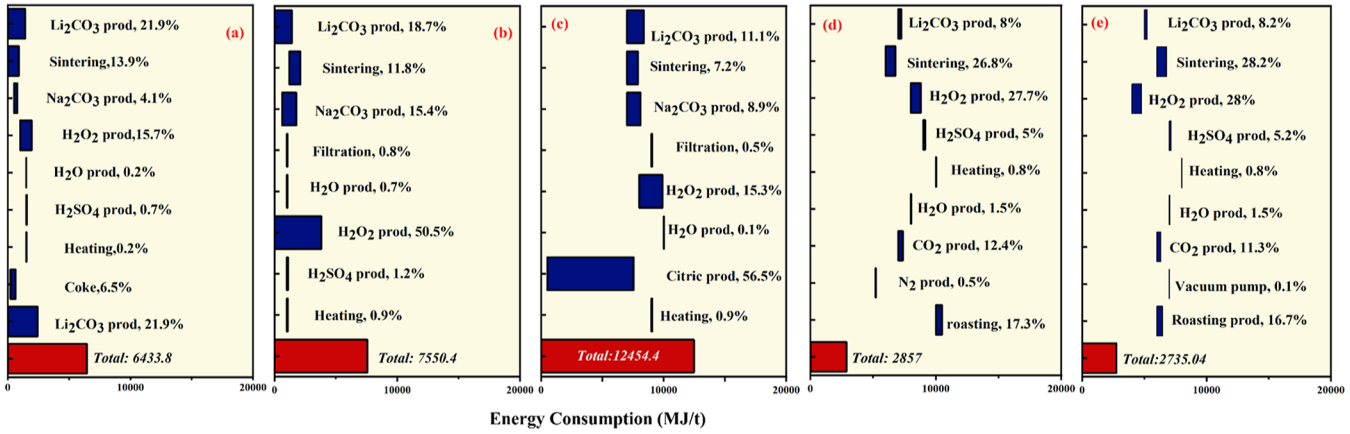


Figure 6. Major contribution of energy consumption in five LIB recycling processes: (a) HTM; (b) HA-SA; (c) HA-CA; (d) IR-VA; (e) IR-N_2.

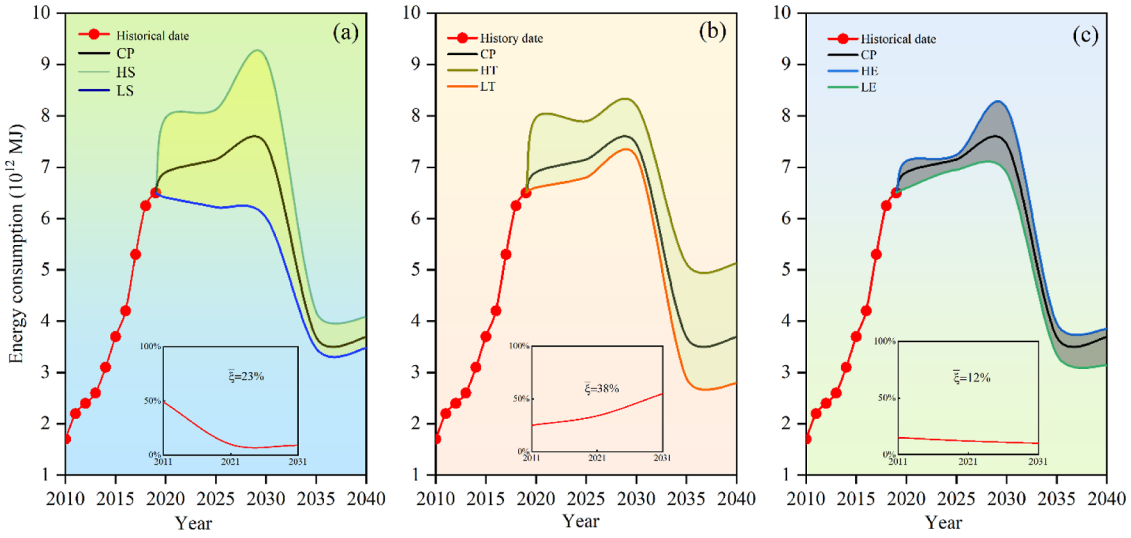


Figure 7. Effects of (a) processing capacity, (b) technology integration, and (c) energy structure on the energy consumption of spent LIB recycling technology.

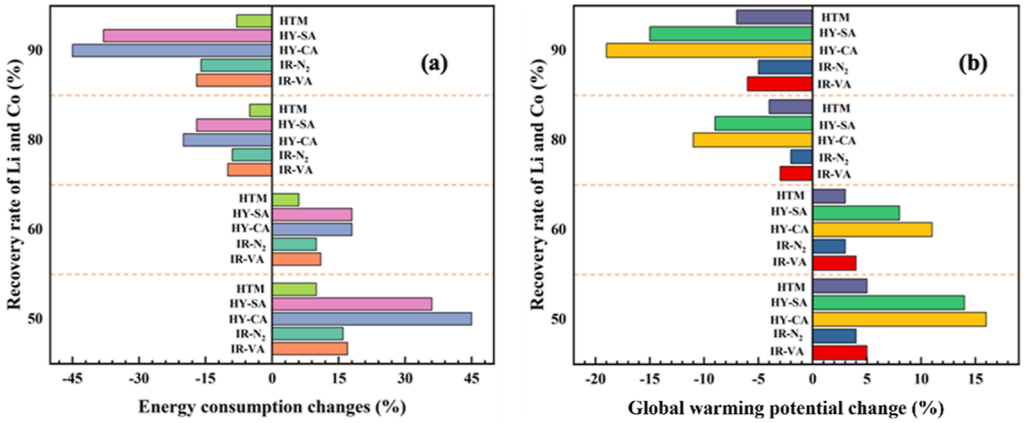


Figure 8. Sensitivity analysis of (a) energy consumption and (b) global warming potential in the five recycling methods by changing the recovery rate of Li and Co.

the total energy consumption. Additionally, this pretreatment processing contributes to 3.6% (HM-CA) to 20.5% (IR-N_2) of the total global warming potential. The automated disassembly of battery packs is currently facing significant challenges due to the wide range of cell types, electrode-active materials, and battery structures employed by different manufacturers in the field of electricity. Thus, it is crucial to establish a standardized approach for batteries in order to effectively streamline future automated dismantling procedures. The current study proposes that the components of LIBs can be effectively separated from the positive and negative active materials. However, in fact, the process of extracting electrode-active materials that are free of

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contamination presents a significant challenge. Following the addition of Cu, Al, Fe, and other impurities, a remediation process will be started. Hence, the inclusion of additional purification methods is imperative in order to attain a homogeneous final product, thereby increasing both the costs and the complexity associated with the overall process.

High-Temperature Method. The HTM faces difficulties associated with increased energy consumption and substantial global warming potential due to its operation at elevated temperatures. The presence of Li in flue fly ash and slag is primarily due to the composition of the electrode material after the smelting process. This poses a significant obstacle to the extraction of Li. As a result, the investment needed for its recovery is excessively high. The low rate of Li in the process can be attributed to the utilization of diverse electrode materials. In situations where LIBs contain a limited amount of valuable metals such as LMO and LFP, the economic viability may be identified as to issue. Umicore is utilized for the direct smelting of spent batteries within a furnace. The methodology employed for data collection exhibits variation. The HTM integrates the pretreatment phase into the LIBs process, facilitating the application of high-temperature smelting to the electrode material power. Consequently, there is a 40% reduction in material handling capacity. It is important to highlight that the electrode material accounts for 60% of the overall weight of the battery. In addition, the utilization of this methodology results in a noteworthy decrease of 20% in the energy consumption, as substantiated by the outcomes of this investigation. The study revealed that the HTM exhibited an energy consumption of 47 MJ (kg recovered-LCO)$^{-1}$, leading to a corresponding global warming potential totaling 2953 kg CO_2-equiv, whereas the Umicore process recorded a higher energy consumption of 54 MJ (kg recovered-LCO)$^{-1}$ and a global warming potential totaling 3393 kg CO_2-equiv. These discoveries highlight the importance of the initial treatment stage in the recovery of spent LIBs and the concentration of valuable components.

Hydrometallurgy Method. Based on the aforementioned investigation of hydrometallurgy, two significant conclusions could be concluded. First, the hydrometallurgy process requires a significant utilization of leaching reagents in order to achieve elevated rates of metal leaching. Nevertheless, this result leads to an increase in the utilization of chemicals and introduces intricacies into the subsequent treatment processes. The utilization of organic acids possesses the capacity to mitigate these concerns to a certain extent. The recycling process requires a higher amount of reagent usage due to the relatively low acidity of organic acids. The significance of life-cycle energy consumption related to tourism production, specifically in relation to these reagents, should not be neglected. A recorded instance of energy consumption in the acid leaching reagent of HM-CA amounted to 9014 MJ, leading to a corresponding global warming potential totaling 392 kg CO_2-equiv. The preceding quantity represented 90% of the aggregate energy consumption throughout the entirety of the conversion process. The production of reagents in HM-SA necessitates an energy input of 4027 MJ, leading to the release of 205 kg of CO_2-equiv in terms of global warming potential. According to the data, the energy consumption related to the manufacturing of chemical reagents in HM-CA is 2.2 times higher than that of HM-SA at this particular stage. Consequently, this leads to the emission of 1.9 times more global warming potential. Thus, based on an analysis of the life cycle impact, it can be concluded that organic acid leaching is an inadequate solution for addressing the environmental concerns associated with inorganic acid leaching. Second, there is a direct correlation between the consumption of reagents and the rate of metal leaching. In order to enhance the extraction of metals, it is crucial to utilize increased quantities of reagents. However, an increase in energy consumption and global warming potential will yield a notably detrimental effect. The sensitivity analysis of the study was also verified. This study analyzes the energy consumption and global warming potential linked to a hydrometallurgy process utilized for metal recovery. The recovery points for the recovery rate of Li and Co were set at 95 and 97%, respectively. Results showed that the total energy consumption exhibited a range of fluctuations, ranging from a reduction of 44% to an augmentation of 45%. Similarly, the corresponding global warming potential demonstrated a variance of approximately 16%.

In Situ Reduction Roasting Method. Compared to alternative techniques, the IR-N_2 approach demonstrated a significantly lower total energy consumption, accounting for just 39% of HTM and 23% of HM-CA. Meanwhile, it was discovered that the corresponding total global warming potential accounted for 68 and 65%. This approach possesses the capacity to not only reduce energy utilization and diminish global warming potential but also restrict the consumption of chemical reagents. However, this approach still poses the challenge of reduced recovery of metallic elements. The present study employed the IR-N_2 process, in which the metal recovery values were determined by referencing the experimental data of pure substances recorded in the existing literature. The calculated percentages Li and Co were determined to be 97 and 98%, respectively. Nonetheless, the present recovery rates for Co and Li in the retrieval of spent LIBs are only 70.43% and 75.08%, respectively.[24] The energy consumption associated with the recycling spent LIBs using LCO is measured at 20 MJ, leading to a corresponding global warming potential of 7 kg CO_2-equiv.[24] It is important to highlight that there is an 11% increase in energy consumption and a 21% increase in global warming potential, which corresponds to a value of ξ_E of 86%. However, this study proposes that the negative graphite undergoes a complete transformation into CO_2, regardless of the presence of transition metals in the elemental state. Besides transition metals, the anticipated metal oxides also pose challenges in the subsequent regeneration stage zone.

Limitations and Recommendation. This study examines five emerging technologies for recycling using spent LIBs and a quantitative analysis to assess the environmental consequences of these technologies throughout their life cycle. The study demonstrates that the five techniques for recycling spent LIBs have substantial potential for minimizing environmental impacts and energy consumption. There are still certain constraints for restrictions in this research. This study focuses on the recovery phase during the life cycle assessment of spent LIBs. However, it is important to note that various technologies are involved in the battery production stage. During this period, the environmental consequences primarily stem from the energy source being used. Hence, it is advisable to do additional research on various manufacturing technologies, diverse energy sources, and various recycling methods during the entire lifespan of spent LIBs. This includes investigating the production, usage, and recovery phases. It is essential to include economic considerations, such as cost, in every stage of the entire battery industry supply chain. In addition, the current research mainly focuses on the recovery of positive valuable metals, while the recovery of other

https://doi.org/10.1021/acssuschemeng.4c04541
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components such as electrolytes, diaphragm, copper aluminum foil, etc. still lacks attention.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.4c04541.

Five recycling technique details; energy consumption change assessment details; results of difference among different types of spent LIBs; major contents in different types of LIBs; summary of sulfuric leaching conditions for recovering Li and Co; summary of citric acid leaching conditions for recovering Li and Co; energy consumption and global warming potential with the production material used in the processes; parameter settings of the technology; and scenario settings (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors are grateful to the reviewers who helped them improve the paper with many pertinent comments and suggestions.

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